واجهات مشروع

تحليل بيانات الرعاية الصحية

HEALTHCARE ANALYTICS

التقنيات المستخدمة

Python

معالجة وتجهيز البيانات

Power BI

بناء الداشبورد والتحليل

A DATA STORY · SAUDI HOSPITALS · 2023 – 2025

01 Overview 02 The Money 03 The Time

Healthcare *Analytics*

تحليل بيانات الرعاية الصحية

THE QUESTION

Where does a hospital's *money* really come from — and where does the patient's *time* go?

من وين يجي دخل المستشفى فعلاً… ووين يروح وقت المريض؟

30,000 rows · 36 months · 6 tables · 30 columns · 120 doctors

ANALYSIS BY **FAY** Data Analytics · Power BI

Patient visits زيارة	**Total billed** إجمالي الفواتير	**Never collected** ما انحصّل	**Average wait** متوسط الانتظار
30,000	SAR 18.1 M	SAR 4.5 M	78 min
across **8 hospitals** 8,000 patients	billed in three years 2023 – 2025	**24.9%** of every riyal billed	**41%** of visits wait 90+ min

Simulated (synthetic) healthcare dataset · for analysis practice

Healthcare *Analytics*

KSA · 2023 – 2025

- Overview نظرة عامة
- **Revenue الإيراد**
- Patient Flow رحلة المريض

ANALYSIS BY

FAY

Simulated healthcare dataset · 30,000 visits · Jan 2023 – Dec 2025

02 · THE MONEY WHERE REVENUE REALLY COMES FROM

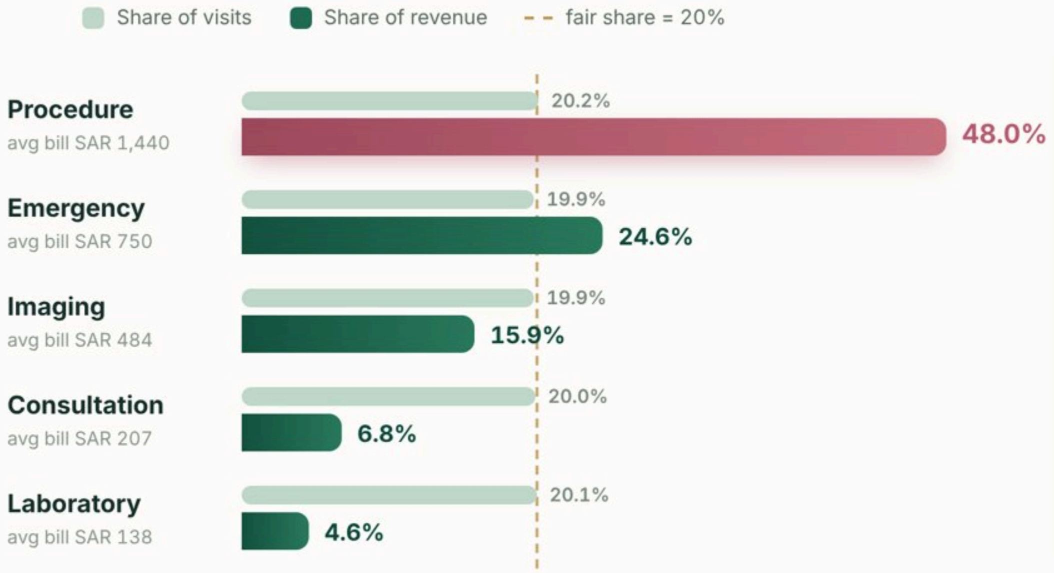
20% of visits bring in *48%* of the revenue

٢٠٪ من الزيارات تجيب نص الإيراد

وربع كل فاتورة ما ينحصّل

Total billed SAR إجمالي الفواتير	Collected SAR المحصّل	Collection rate نسبة التحصيل	Never collected SAR ما انحصّل	Average bill SAR متوسط الفاتورة
18.14M	13.62M	75.1%	4.52M	605
▲ 1.8% vs 2024	▲ 1.6% vs 2024	▼ 0.2 pts vs 2024	▲ 2.5% vs 2024	▲ 1.5% vs 2024

Share of visits vs share of revenue الزيارات مقابل الإيراد



48%

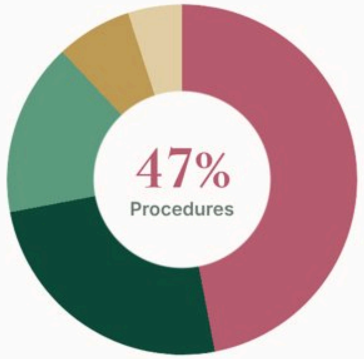
of all revenue comes from **Procedures**, only 20% of visits.

×10

a procedure bills **SAR 1,440** on average; a lab test bills SAR 138.

Where the SAR 4.5M leaks وين يضيع التحصيل

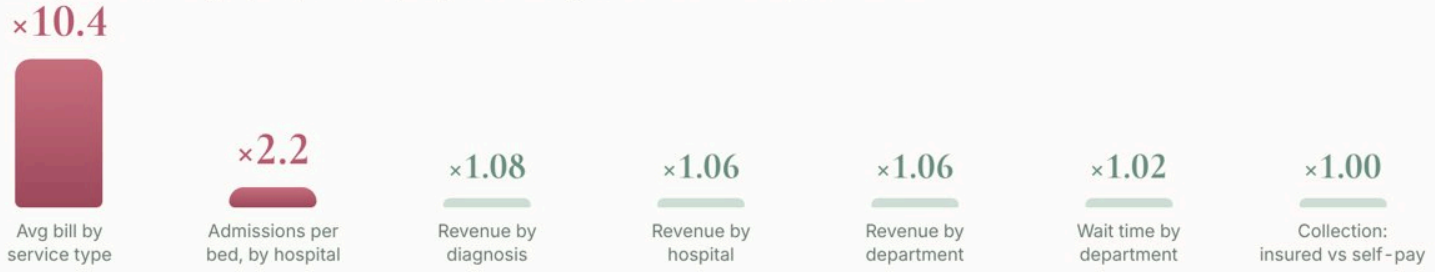
Billed minus collected, by service



🟥 Procedure	SAR 2.14M	47%
● Emergency	SAR 1.14M	25%
● Imaging	SAR 0.72M	16%
● Consultation	SAR 0.31M	7%
● Laboratory	SAR 0.21M	5%

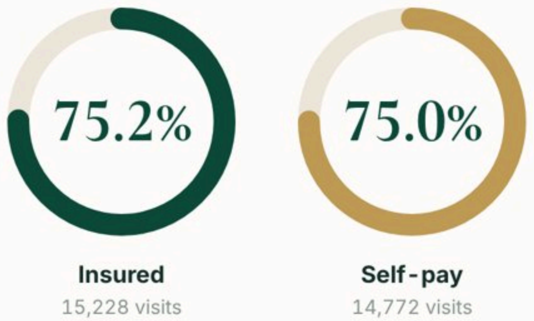
Most things are flat. Two aren't. وين الفرق الحقيقي؟

Highest ÷ lowest group. Departments, hospitals and diagnoses all sit within 8% of each other.



Insurance doesn't speed up payment

التأمين ما فرق بالتحصيل



0 of 30,000 bills paid in full

1 in 5 collected under 60%

Healthcare *Analytics*

KSA · 2023 – 2025

Overview
نظرة عامة

Revenue
الإيراد

Patient Flow
رحلة المريض

03 · THE TIME WAITING, BEDS AND OUTCOMES

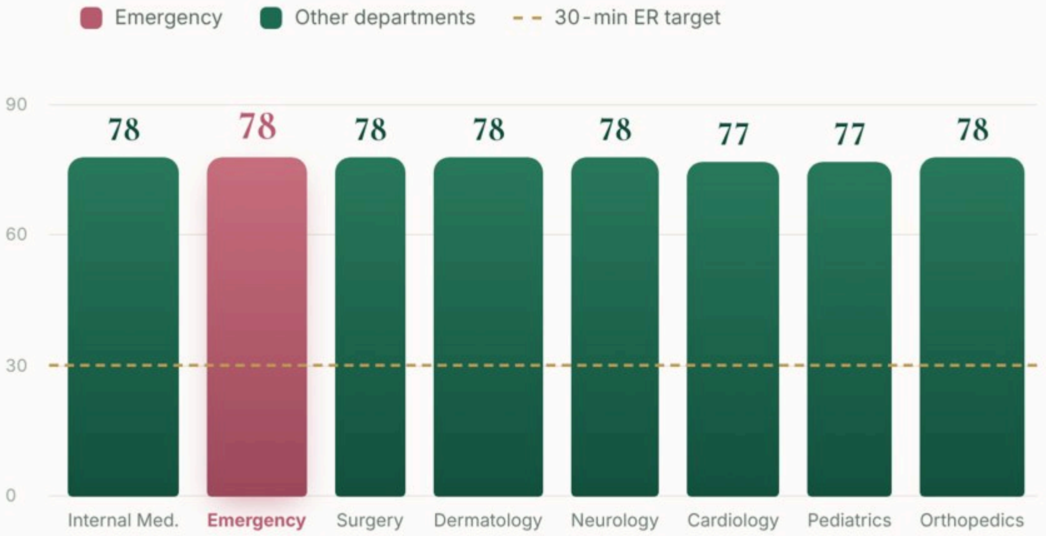
An ER patient waits as long as *everyone else*

مريض الطوارئ ينتظر مثل الكل

٧٨ دقيقة بكل الأقسام… والمشكلة بالنظام كله

Average wait متوسط الانتظار	77.9 min	▼ 0.7% vs 2024
Waited 90+ min فوق ٩٠ دقيقة	41.4%	▼ 1.1 pts vs 2024
ER seen in 30 min الطوارئ خلال ٣٠ د	17.8%	▲ 0.1 pts vs 2024
Admission rate نسبة التنويم	33.4%	▲ 0.8 pts vs 2024
Referred out نسبة التحويل	33.3%	▼ 0.6 pts vs 2024

Average wait by department (minutes) متوسط الانتظار حسب القسم



78 min

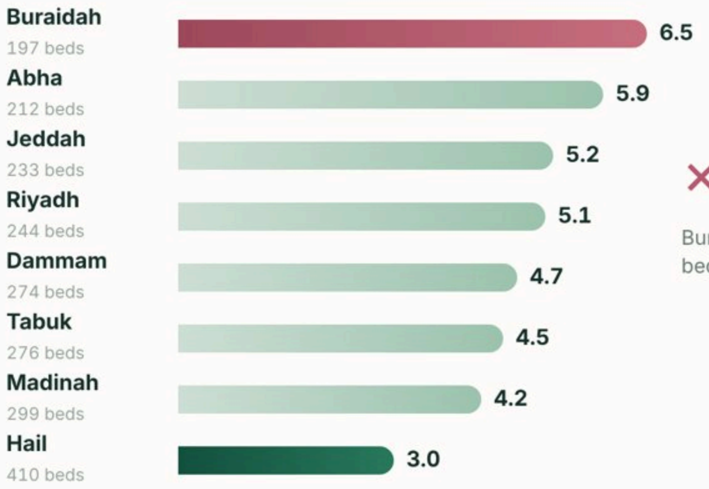
average **ER** wait, the same as Dermatology (78 min).

18%

of ER visits are seen within 30 min, no better than all visits (18%).

Same demand, very different beds ضغط الأسرّة

Admissions per bed · every hospital sees ~3,700 visits

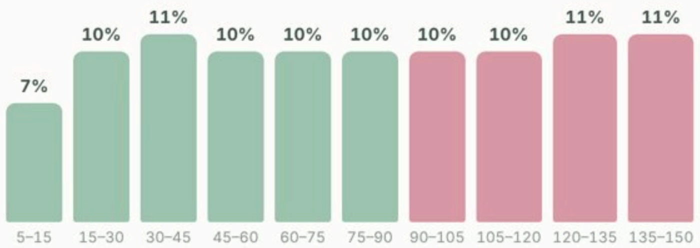


×2.2

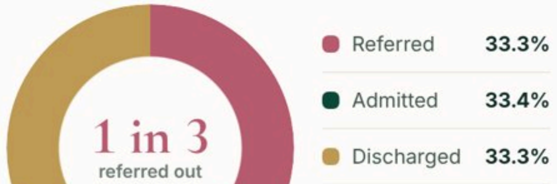
Buraidah's load per bed vs Hail's

No fast lane ما فيه مسار سريع

Share of visits by wait (minutes) · spread evenly



Where visits end نتيجة الزيارة



9,997 visits referred elsewhere: care, time and revenue leaving.

What should we do? وش نسوي؟

1. **Give the ER a real fast lane**
   ER waits 78 min like everyone; only **18%** are seen in 30 min. Triage first, track it monthly.
2. **Chase the SAR 4.5M, procedures first**
   Procedures hold **47%** of the gap (SAR 2.14M). Deposits and pre-approvals before the procedure.
3. **Balance the beds**
   Buraidah carries **2.2×** Hail's admissions per bed. Route planned admissions to Hail and Madinah.

ANALYSIS BY

FAY

Simulated healthcare dataset · 30,000 visits · Jan 2023 – Dec 2025